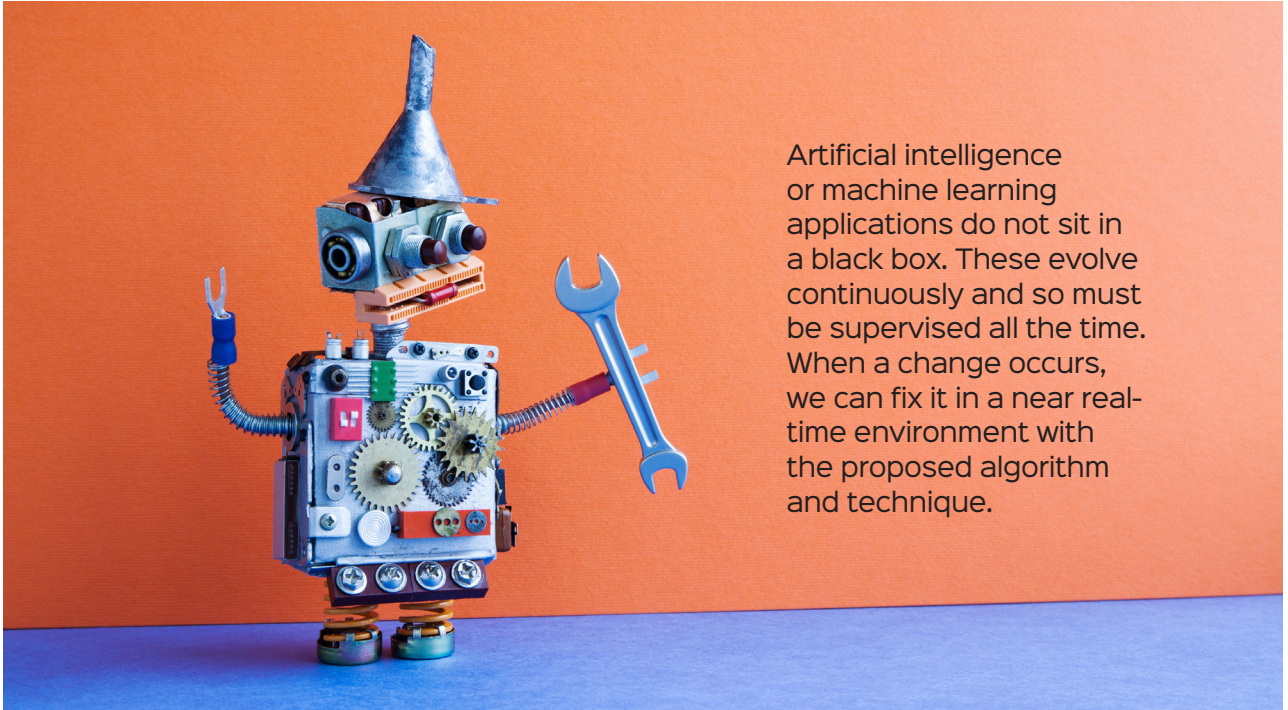


AI: The Curious Case of Concept Drift



Artificial intelligence or machine learning applications do not sit in a black box. These evolve continuously and so must be supervised all the time. When a change occurs, we can fix it in a near real-time environment with the proposed algorithm and technique.

Think about a scenario where you are using an electric motor on your shop-floor. You have installed a few sensors on this motor's outer body, and these sensors are continuously sending the data over the wireless network. Moreover, you also have an elaborate setup on the cloud where your application analyses these parameters and determines your motor's health status.

Of course, this scenario considers the trained pattern of motor vibration and current consumption by the motor during the learning phase. Moreover, so long as nothing changes, this pattern recognition works like a charm. That is what a good machine learning outcome would look like.

However, the data generated by an electric motor can change over time.

It can result in poor analytical results, which otherwise assumes a static relationship between key parameters and motor-health.

The change in data occurs due to various real-life scenarios. These scenarios range from changes in operating load conditions, aging of mechanical components such as ball-bearings, or wear and tear of the foundation on which the motor is installed. Environmental conditions can change, and several other factors may get affected too.

Nonetheless, this occurrence is quite common for several other real-life scenarios of machine learning. The data changes over time and affects statically assumed and so programmed relationships. In technical jargon, it is known as 'concept drift.' The word concept

refers to an unknown and hidden relationship between the output and its input variables. These inputs are also known as 'hidden context.'

What is the problem here?

For a static use case, concept drift does not pose any problem at all. However, in most use cases, the relationship between input parameters, that is, features and output characteristics, changes with time. If you have assumed this relation to be static during the machine learning model development, it could be an issue in the future.

This problem may be relatively easier to handle if you are maintaining these relationships and formulae on the cloud. You can update the new formulae and relationships, and everything will be back to normal.